Replication: Wu (2018) AEA

Keegan Fitzpatrick
University of Wisconsin-Madison

February 11, 2026

1 Paper Overview

Wu (2018) examines gendered language patterns in the Economics Job Market Rumors (EJMR) forum, which suggests evidence of an unwelcoming or stereotypical culture in economics towards women and other minority groups. Using a Lasso-Logistic regression model, Wu (2018) identifies words most strongly associated with discussions about women versus men in anonymous forum posts. The analysis revealed that the words most predictive of the posts about women typically focused on physical appearance and personal information such as family, while the words most predictive of the posts about men tended to emphasize academic and professional characteristics. The study analyzed over 2 million EJMR posts from October 2013 to October 2017, identifying over 400,000 gendered posts based on the classification scheme. The findings suggested systematic differences in how women and men are portrayed in anonymous professional discussions, highlighting the potential gender biases present in the economics profession.

2 Replication Description

Replication resources for Wu (2018) can be found here: [1]. This replication package requires downloads of all files present on the above webpage. The 3 Python scripts are out of date, using old NumPy functions, which have since been rendered obsolete. With this in mind, individuals who wish to replicate the original paper for themselves should download the modified Python scripts found here: [2]. These scripts run off of large datasets and, as a result, will take several minutes to complete, depending on CPU speed. One might also take advantage of my modified .R file, which auto-saves PDF versions of relevant tables and figures, including the Random Forest extension.

3 Replication Results

Wu (2018) uses LASSO logistic regression to identify the words most predictive of a poster's gender status on EJMR. Letting $\mathbf{w}_i$ denote a vector of word counts for post i (excluding gender classifiers), Wu estimates a LASSO-regularized logistic model for the probability that the post is about a woman:

$$\hat{\theta}_\lambda = \arg\min_{\theta} - \log\left(\prod_{i=1}^{N} P(Female_i|\mathbf{w}_i)\right) + \lambda||\theta||_1$$

where $||\theta||_1 = \sum_{j\geq 1} |\theta^j|$. The LASSO regularization (L1 penalty term $\lambda||\theta||_1$) selects the most relevant features from 9,540 words by setting less important coefficients to exactly zero.

Table 1 presents the top 10 words most predictive of female and male posts in the full sample. Words strongly associated with female posts focus predominantly on physical appearance and personal characteristics: "hotter" (ME = 0.422), "pregnant" (0.323), "attractive" (0.245), and "beautiful" (0.240). In contrast, words associated with male posts emphasize professional and academic attributes: "homo" (-0.303), "testosterone" (-0.195), "macroeconomics" (-0.180), and "chapters" (-0.189). These marginal effects (ME) can be interpreted, for example, as each occurrence of "hotter" increases the log-odds of female classification by 0.422. Table 2 replicates this analysis using only the pronoun sample (posts containing explicit gendered pronouns) and finds basically an identical pattern, showing consistency across more restricted samples. Figure 1 tracks the fraction of female and male posts containing any of their respective top 50 predictive words over time from 2013–2017. The results of Wu (2018) provide compelling evidence that anonymous discussions about women in economics systematically emphasize appearance and personal life, while discussions about men focus on professional qualifications and academic work, revealing gender stereotypes embedded in professional discourse.

word	ME_pronoun	word	ME_pronoun
pregnancy	0.2922947	knocking	−0.3294384
hotter	0.2893404	testosterone	−0.2043009
pregnant	0.2583185	blog	−0.1825698
hp	0.2380338	hateukbro	−0.1764642
vagina	0.2280140	adviser	−0.1745582
breast	0.2201753	hero	−0.1738850
plow	0.2190030	cuny	−0.1732126
shopping	0.2073728	handsome	−0.1661303
marry	0.2072572	mod	−0.1660732
gorgeous	0.2008608	homo	−0.1599676

word	ME_pronoun	word	ME_pronoun
pregnancy	0.2922947	knocking	−0.3294384
hotter	0.2893404	testosterone	−0.2043009
pregnant	0.2583185	blog	−0.1825698
hp	0.2380338	hateukbro	−0.1764642
vagina	0.2280140	adviser	−0.1745582
breast	0.2201753	hero	−0.1738850
plow	0.2190030	cuny	−0.1732126
shopping	0.2073728	handsome	−0.1661303
marry	0.2072572	mod	−0.1660732
gorgeous	0.2008608	homo	−0.1599676

Figure 1: Left: Table 1 - Top 10 Words Most Predictive of Female/Male Posts (Full Sample). Right: Table 2 - Top 10 Words Most Predictive of Female/Male Posts (Pronoun Sample)

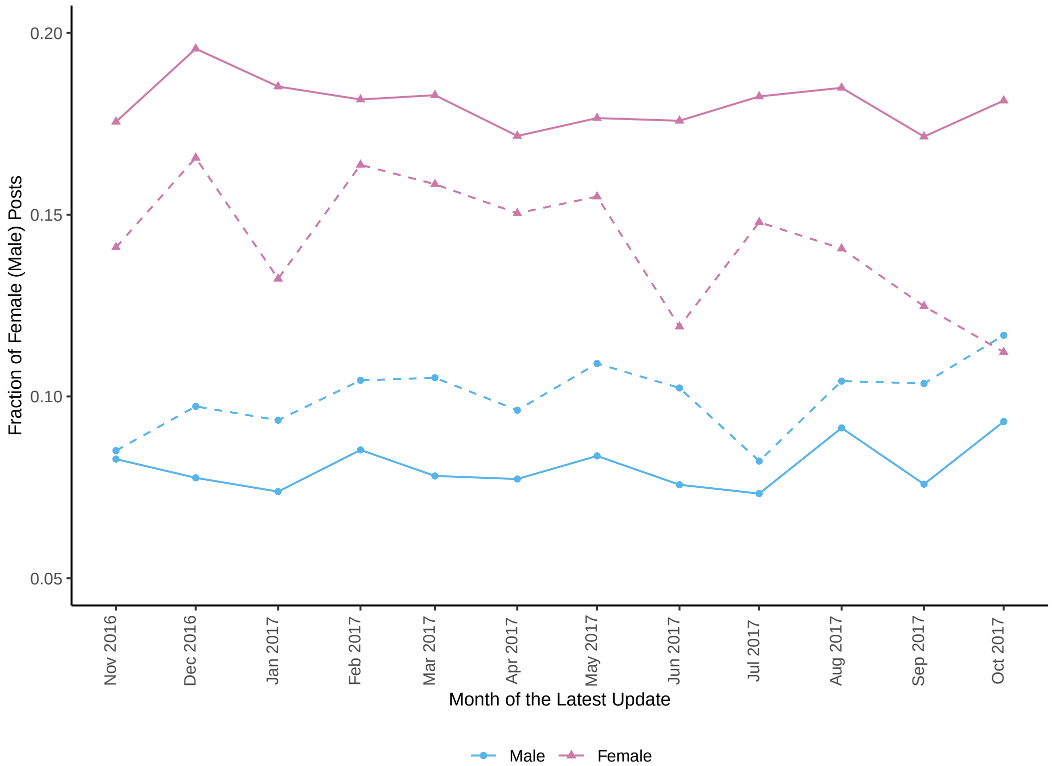

Figure 2: Figure 1: Fraction of Female (Male) Posts Containing Top 50 Predictive Words Over Time

4 Random Forest Model

Overview

This extension employs a Random Forest classifier to analyze gendered language patterns as a complement to Wu (2018)'s LASSO logistic regression approach. While Wu's linear model assumes that each word contributes independently and additively to predicting post gender, Random Forests can capture non-linear relationships and interactions between words that a linear model cannot detect. For example, LASSO may assume "assistant" has the same effect regardless of context, while Random Forests might learn that "assistant" in combination with "professor" has a different gender association than "assistant" alone.

The Random Forest algorithm works by constructing 100 independent decision trees, each trained on a random sample of the scraped posts, so each tree sees a slightly different cluster from the next. At each node in a decision tree, the algorithm considers a random subset of approximately 98 words (out of the 9,540 total) and selects the word that best separates male and female posts based on what is known as a Gini impurity criterion. The final prediction for any given post is simply determined by the majority vote across all 100 trees. The Random Forest results demonstrate results similar to the original model, identifying many of the same words.

Limitations and Comparison

The Random Forest model suggests that Wu (2018)'s choice of LASSO logistic regression was methodologically correct. While Random Forest can capture non-linear patterns and interactions, the overlap in top predictive words and similar overall patterns suggest that gendered language in EJMR forum posts is primarily linear and additive. The interpretability benefits of LASSO coefficient interpretations and explicit feature selection outweigh more complex methods. Wu (2018)'s model appears to sacrifice little in accuracy while maintaining greater interpretability of results compared to Random Forest.

The interpretability trade-offs can be summarized by the fact that LASSO uses a single equation where each word has a fixed coefficient that directly tells you its effect size and direction. Random Forest makes predictions by majority vote across 100 decision trees, each with different hierarchical splitting rules (which are based on feature selection). The "importance" of a word measures how useful it was for creating splits across all trees, but this does not translate to a fixed effect size like LASSO. You cannot trace a single prediction back to specific word contributions because it emerges from the collective decision of 100 different decision paths.

Table 3 shows the top 10 words identified by the Random Forest model. The results largely confirm Wu's findings: female-associated words focus on physical appearance ("hot," "attractive," "beautiful") and personal relationships ("marry," "sex," "date"), while male-associated words emphasize academic terminology ("macro," "economics," "theory," "paper"). The substantial overlap with LASSO results validates Wu (2018)'s methodological approach and demonstrates that gendered language patterns are robust across different modeling frameworks.

word	RF Importance	word	RF Importance
hot	0.0484104	of	–0.0151639
marry	0.0301445	the	–0.0142151
sex	0.0289417	macro	–0.0104032
your	0.0236584	is	–0.0096760
date	0.0219655	,	–0.0081131
attractive	0.0210934	top	–0.0072127
beautiful	0.0186138	0	–0.0071578
kids	0.0170159	economics	–0.0070953
my	0.0167467	theory	–0.0069005
married	0.0162810	paper	–0.0061160

Figure 3: Table 3: Top 10 Words Most Predictive in Random Forest Classifier

References

1. Wu, A. (2018). Gendered language on the economics job market rumors forum. *AEA Papers and Proceedings*, 108, 175-179.

2. Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32.